V-Model (Verification & Validation Model)

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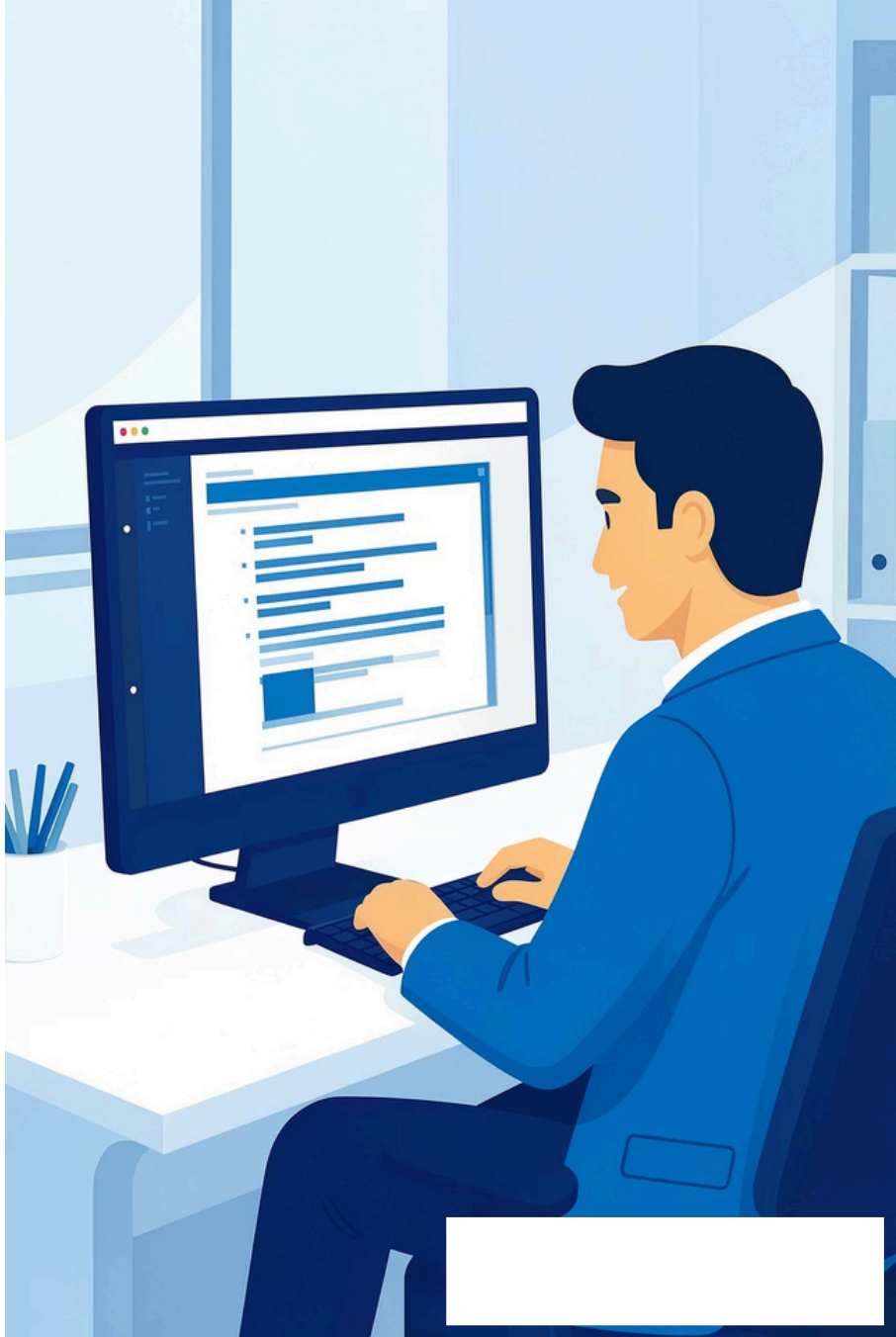
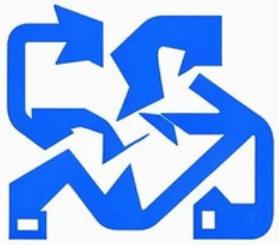
Introduction

- Definition: A sequentialdevelopment model emphasizing paired verification and validation activities.
- Designed to overcome key limitations of the Waterfall model.
- Emphasises early and continuous testing aligned with development phas es.
- Improves overall software quality and traceability.



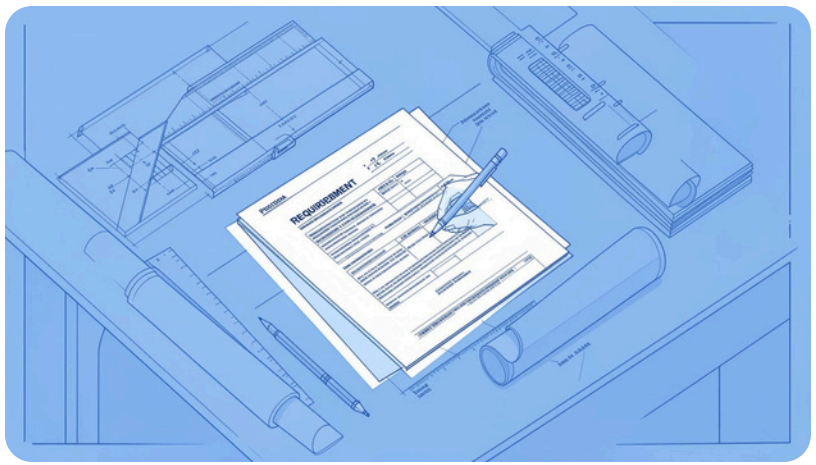


Why the V-Model?

- Problems withWaterfall:latediscovery of defects and prolonged feedback loops.
- Late testing: testing mostly after implementation.
- High defect fixing cost when issues are found late.
- Necessitates early testing and verification to reduce risk.

Structure of the V-Model

- V-shapedlifecycle:development on left,codingat bottom, testing on right.
- Left side (Verification): planning and design activities.
- Bottom(Coding) : implementation.
- Right side (Validation): unit → integration → system → acceptance testing.

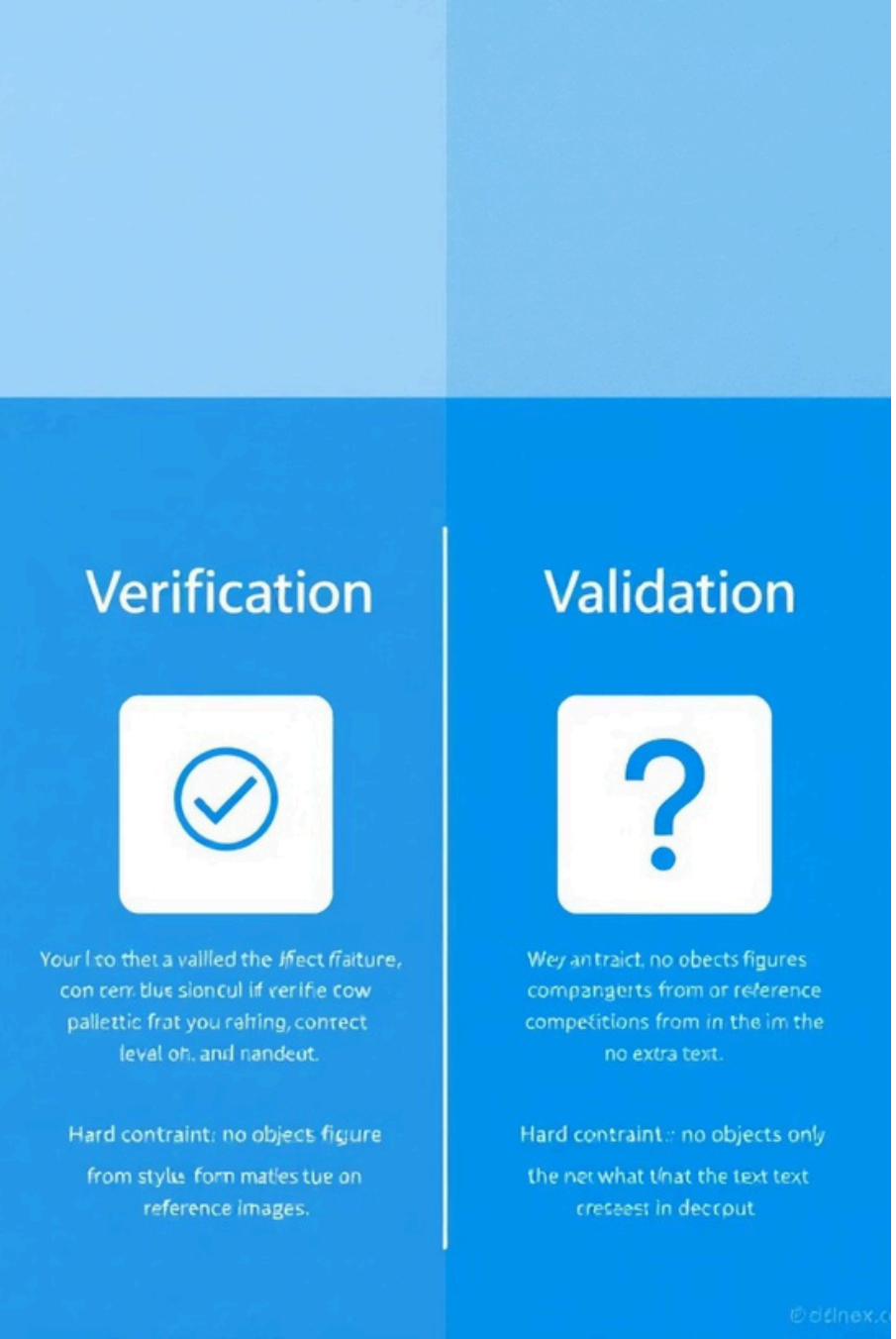


Verification Phase (Left side)

- RequirementAnalysis: capture functional& non-functional needs.
- System Design: define system architecture and interfaces.
- High-Level Design (HLD): module boundaries and data flow.
- Low-Level Design (LLD): detailed algorithms and data structures.
- Goal: Build the product *correctly*.

Coding & Validation Phase (Right side)

- Coding: translate LLD into executable code.
- Unit Testing: verify individual components (LLD ↔ Unit).
- Integration Testing: verify interfaces and combined modules (HLD ↔ Integration).
- System & Acceptance Testing: validate full system against requirements.



Verification vs Validation

- Verification:"Arewebuildingtheproductcorrectly?" — checks artifacts, designs.
- Validation: "Are we building the right product?" — checks behaviour against requirements.
- Phase mappings:
- Requirement Analysis ↔ Acceptance Testing
- System Design ↔ System Testing
- High-Level Design ↔ Integration Testing
- Low-Level Design ↔ Unit Testing

Advantages

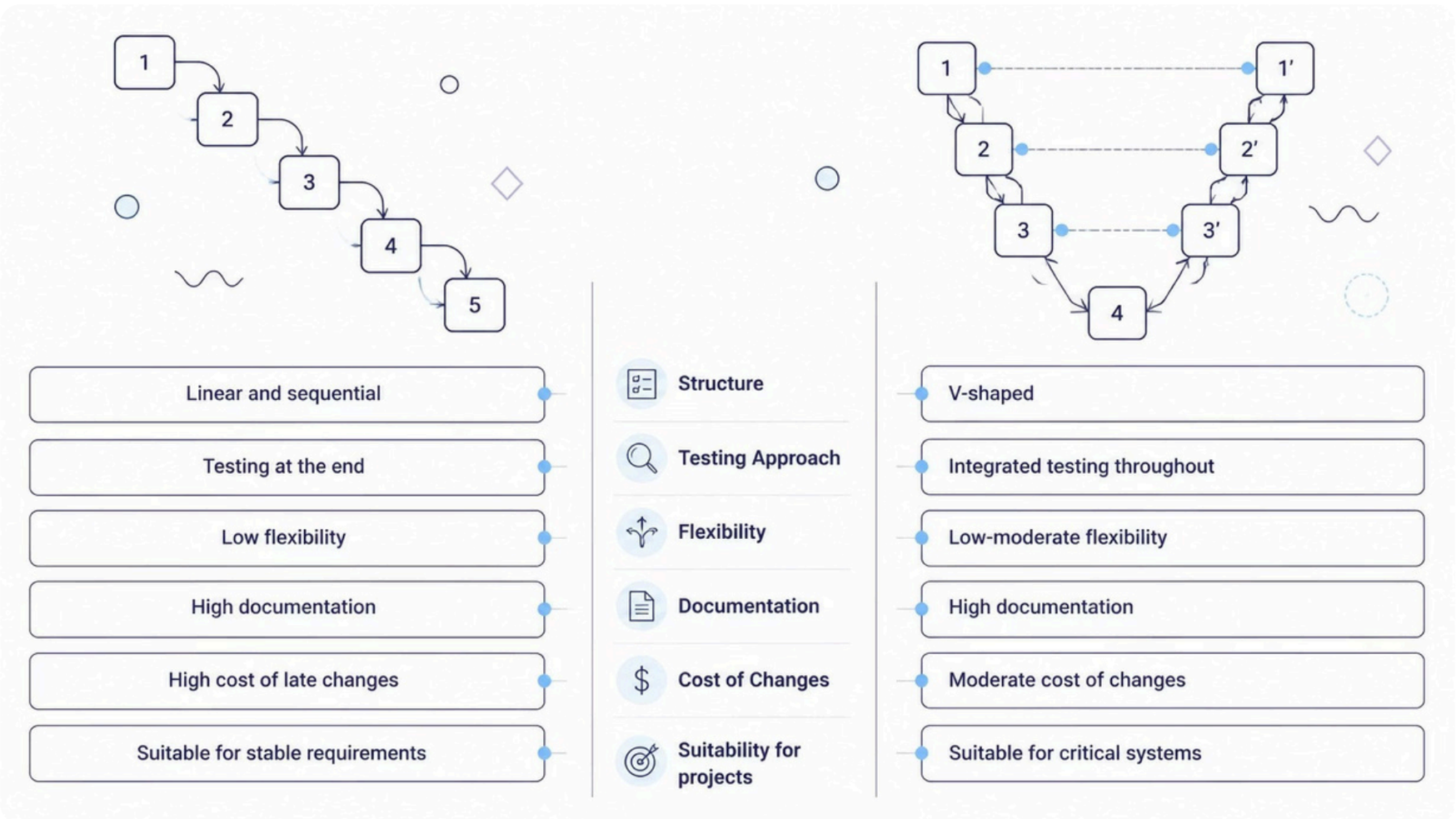
- Earlydefectdetection: verification begins with requirements.
- Improved software quality through parallel test planning.
- Reduced development cost by catching issues early.
- Better planning and clearer deliverables per phase.
- Easy project tracking and high reliability for critical systems.

Limitations

- Notsuitableforchanging requirements: inflexible to change.
- Rigid process: strict phase-to-phase flow.
- High documentation overhead for each verification level.
- Expensive for small projects and no prototyping by default.

Waterfall vs. V-Model: A Comparative Overview

Understandingthekey differencesbetweentheWaterfallandV-Modelhelpsinselectingtherightapproach for softwaredevelopment basedonprojectneedsandrisktolerance.



While both models are sequential, the V-Model's integrated testing approach significantly improves defect detection and reduces the impact of changes compared to the traditional Waterfall approach.

Conclusion: Key Takeaways

Integrated Quality Assurance

TheV-Model seamlessly integratestesting with development from initial requirements, moving beyond Waterfall's late-stage testing.

Early Defect Detection

Proactive verification and validation phases ensure issues are discovered and resolved significantly earlier, reducing overall project costs.

Enhanced Traceability

ItsV-shapedstructure provides clear, direct mappings between each development activity and its corresponding testing phase, ensuring comprehensive coverage.

Ideal for Critical Systems

Withitsrigorous,structuredapproach, the V-Model is particularly well-suited for projects requiring high reliability and stable, well-defined requirements.

Choosing the V-Model strategically empowers teams to deliver high-quality, reliable software, particularly in environments where precision and early validation are paramount.

Thank You

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